

Fathima Nafra

BSc (Hons) in Computer Science (Specialization: Data Science) - Undergraduate

 www.linkedin.com/in/fathimanafra |  <https://github.com/FathimaNafra>

 nafrafathi.me@gmail.com |  077-9150212 |  Sri Lanka

<https://www.nafra.me/>

Profile Summary

Computer Science undergraduate at the University of Kelaniya, Faculty of Computing & Technology with a strong interest in data analysis, data engineering, and machine learning. Experienced in collecting, processing, and analyzing data to support real-world decision-making. Skilled in working with datasets, building data workflows, and applying machine learning techniques to uncover patterns and insights. Motivated to continuously develop technical and analytical skills, with a goal of contributing to practical, data-driven solutions.

Education

BSc. (Hons) in Computer Science (Specialization: Data Science) 2023 - Present
Faculty of Computing & Technology, University of Kelaniya
Current GPA: 3.62/4.00

Diploma in ICT & English 2022-2023
ICBT Galle Campus

G.C.E. Advanced Level - Physical Science Stream Completed: 2021
Results: Physics - B, Chemistry - B, Combined Mathematics - A
Rippon Girls' College, Galle

Projects

- 1. Loan Default Risk Approval Prediction System** 2026
 - Developed LightGBM classification & regression models to predict loan approval and generate risk scores.
 - Improved model performance through preprocessing, feature engineering, and hyperparameter tuning.
 - Applied SHAP for model interpretability and feature importance analysis.
 - Built a Flask web app for real-time prediction and user interaction.
 - *Technologies:* Python, LightGBM, SHAP, scikit-learn, Flask
- 2. End-to-End Azure Data Engineering Pipeline** 2026
 - Designed and implemented a scalable data pipeline with automated ingestion into Azure SQL.
 - Structured data processing using Medallion architecture (Bronze - Silver layers).
 - Performed efficient data transformation and processing using PySpark in Databricks.
 - *Technologies:* Azure Data Factory, Azure SQL, Databricks, PySpark
- 3. Finance Lakehouse Data Engineering Project** 2026
 - Built an end-to-end Lakehouse data pipeline in Databricks using synthetic financial transaction data generated for analytics and data engineering workflows.
 - Implemented Medallion Architecture (Bronze, Silver, Gold) using Delta Lake to support reliable and scalable data processing.
 - Developed PySpark ETL pipelines for data ingestion, transformation, fraud detection, and KPI generation.

- Designed interactive analytics dashboards to monitor revenue performance, transaction trends, customer behavior, and fraud risk metrics.
- *Technologies:* Databricks, Apache Spark, PySpark, Delta Lake, SQL, Databricks SQL Dashboards, GitHub

4. AI-Based Crypto & Forex Signal Generator

2026

- Built a full-stack app generating real-time trading signals using RSI, EMA, MACD, and ML models.
- Developed backend APIs with Flask and frontend UI using React (Vite).
- *Technologies:* Python, Flask, React, Pandas, Scikit-learn

Collaborated in a team of 2 to design and develop the system, contributing to both frontend and backend development.

5. HallEase - Event & Hall Management System

2nd year - Group Project

- Developed a web-based system for hall booking, scheduling, and admin management.
- Improved booking efficiency and reduced scheduling conflicts.
- *Technologies:* TypeScript, Next.js, Tailwind CSS, Supabase

6. Netflix Movies & TV Shows Data Analysis

2026

- Analyzed Netflix dataset using SQL to extract insights on trends, genres, and distribution.
- Built Power BI dashboard to visualize KPIs and content patterns.
- *Technologies:* SQL, Power BI

Certifications

- Azure Fundamentals Course for Beginners - ScholarHat
- Machine Learning with Python Free Course - ScholarHat
- AWS AI Practitioner Challenge - Udacity
- Power BI for Beginners - Simplilearn
- SQL Server Foundations Course - ScholarHat
- Python 101 for Data Science - Cognitive Class
- AWS Certified Machine Learning Engineer - Associate - Udemmy (Reading)

Technical Skills

- **Programming Languages:** Python, R, JavaScript
- **Web Development & Frameworks:** React, Flask, HTML, CSS
- **Machine Learning & Data Analysis:** scikit-learn, pandas, NumPy, Matplotlib, Seaborn, Streamlit
- **Data Engineering & Big Data:** ETL pipelines, Azure Data Factory, Azure Databricks, PySpark
- **Databases:** MySQL, Azure SQL, PostgreSQL, Supabase, PL/SQL
- **Cloud & Development Tools:** Git, GitHub, Azure, AWS (basic), Jupyter Notebook, Power BI

Soft Skills

Analytical Thinking | Problem Solving | Clear Communication | Adaptability | Cross-Functional Collaboration